

Multiple Choice (10 marks)

1. Approximately how close is the Andromeda Galaxy?
 - (a) 1.7 million light years
 - (b) 2.1 million light years
 - (c) 2.5 million light years
 - (d) 3.2 million light years
2. What Mars mission will be landing on July 4 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) ExoMars
 - (b) Mars Exploration Rovers
 - (c) Mars Science Laboratory
 - (d) Phoenix Mars Lander
3. Find the best approximation for the core temperature of the Sun:
 - (a) 6000 K
 - (b) 7000 K
 - (c) 9000 K
 - (d) 13000 K
4. Calculate the ratio of the solar radiation flux on Mercury's surface for aphelion (0.304 AU) versus aphelion (0.456 AU).
 - (a) 4:1
 - (b) 1:2
 - (c) 6:5
 - (d) 9:4
5. Why are neutron stars relevant for astronomers?
 - (a) To measure interstellar mass.
 - (b) To measure galactic distances.
 - (c) To measure galactic energy-density.
 - (d) To measure interstellar density.

True / False (10 marks)

Answer True or False

6. True or False: Titan is the only outer solar system moon that has methane lakes on its surface, indicating potential habitability.
7. True or False: Planetary rings can be formed from the accumulation of dust by impacts over time.
8. True or False: The black hole at the center of the Milky Way is referred to as Sagittarius A*.
9. True or False: Kirkwood gaps occur in the main asteroid belt at positions where asteroids orbit with a period resonant with Jupiter.
10. True or False: The belts and zones of Saturn are alternating bands of rising and falling air at different latitudes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of **helium** as the most abundant element among the Jovian planets and analyze its implications for their formation and atmospheric characteristics.
12. Calculate the mass of a spherical moon with a diameter of 150 km composed entirely of water ice, and analyze the ratio of this mass to that of Saturn's rings, which is 2×10^{19} kg. Discuss the implications of this ratio in the context of celestial body composition and formation.

End of Paper